

Ex.no: 11

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NETWORK ADDRESS TRANSLATION (NAT) USING CISCO PACKET TRACER

Aim:

To implement Network Address Translation (NAT) in Cisco Packet Tracer and allow private devices to communicate with an outside network.

Procedure:

1) Add devices

- 2 PCs
- 1 Switch
- 1 Router
- 1 Cloud / Another Router

2) Connect devices

- PC1, PC2 → Switch
- Switch → Router G0/0
- Router G0/1 → Outside Router / Cloud

Assign IP address:

Device	IP address	Gateway
PC1	192.168.1.10	192.168.1.1
PC2	192.168.1.11	192.168.1.1
Router G0/0	192.168.1.1	—
Router G0/1	200.1.1.1	—

Configure router

Open router → CLI and enter
enable

configure terminal

interface gigabitEthernet 0/0

ip 192.168.1.1 255.255.255.0

ip nat inside

no shutdown

exit

interface gigabitEthernet 0/1

ip 200.1.1.1 255.255.255.0

ip nat outside

no shutdown

exit

Configure static NAT:

```
ip nat inside source static 192.168.1.10 200.1.1.10
```

Configure dynamic NAT:

```
ip nat pool My_pool 200.1.1.20 200.1.1.25
```

```
netmask 255.255.255.0
```

```
access-list 1 permit 192.168.1.0 0.0.0.255
```

```
ip nat inside source list 1 pool My_pool
```

Test NAT

From PC1, open command prompt and type:

```
ping 200.1.1.2
```

Check NAT translations on the router. Show IP NAT translations to verify that the translation has occurred.

Result:

NAT was successfully configured and private IP addresses were translated to public IP addresses for communication with the outside network.